Chia-Chi (Carol) Hsu

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Education

National Tsing Hua University (NTHU), Hsinchu, Taiwan Sept. 2023–Dec. 2025
M.A., Special Education (GPA: 4.30/4.30)
Thesis: *Cerebral Lateralization of Language and Face Processing in Children with Autism Spectrum Disorder*
Advisor: Prof. Hsin-Jen Hsu

University of California, Berkeley, CA, USA Aug. 2024–May 2025
Visiting Research Student, Berkeley International Study Program (BISP)
Ministry of Education Academic Talent Development Fellow

National Tsing Hua University (NTHU), Hsinchu, Taiwan Sept. 2019–May 2023
B.A., Special Education (GPA: 4.14/4.30)

Research Interests

Developmental cognitive neuroscience; neural mechanisms of language and reading development in children; individual differences in brain–behavior relationships; multimodal approaches to studying brain development and learning (fTCD, EEG/ERP, fMRI,).

Publications & Manuscripts

Hsu, C.-C., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F. (2026). Baseline mathematical aptitude attenuates group differences in posterior P300 during arithmetic verification in high school athletes. *Experimental Brain Research, 244*, 163. https://doi.org/10.1007/s00221-026-07366-y
Hsu, C.-C., Yu, H.-Y., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F. Posterior P300 differences are more robust during multiplication than function verification. *Manuscript in preparation.*
Hsu, C.-C., & Yao, Z.-F. The Taiwanese blueprint for inclusive aquatics for children with medical dependencies. *Manuscript in preparation.*

Conference Presentations

Hsu, C.-C., & Hsu, H.-J. (2026, November). *Atypical lateralization of word processing but not face processing in children with autism spectrum disorder*. Poster presentation, Psychonomic Society Annual Meeting, San Diego, CA.
Hsu, C.-C., Yu, H.-Y., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F. (2026, November). *Task-invariant P300 responses in athletes during mathematical tasks*. Poster presentation, Psychonomic Society Annual Meeting, San Diego, CA.
Hsu, C.-C., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F. (2026, June). *Enhanced posterior P300 during arithmetic verification reflects differences in mathematical proficiency*. Poster presentation, Summit of Psychological Associations in the Asia-Pacific (APPA), Tainan, Taiwan.
Hsu, C.-C. (2025, November). *Exploring language proficiency and anxiety among Mandarin-speaking exchange students at UC Berkeley*. Poster presentation, International Conference on Educational Innovation (ICEI), Hsinchu, Taiwan
Hsu, C.-C., & Hsu, H.-J. (2023, December). *Brain lateralization of language and visual processing in individuals with and without ASD*. Poster presentation, Asia Pacific Society of Speech, Language and Hearing (APSSLH), Ho Chi Minh City, Vietnam.

Research Experience

Full-time Research Assistant — Agile Brain and Cognitive Lab, NTHU 2026–Present
Advisor: Prof. Zai-Fu Yao
Project: *Memory Encoding and Exercise Intervention*

- Conduct structural and functional MRI analyses using FreeSurfer, FSL, and SPM, including cortical reconstruction, volumetric analyses, preprocessing, and first-level modeling.
- Examine whether exercise-related changes in hippocampal and prefrontal activation are associated with individual differences in memory performance and learning outcomes.

Project: *Mathematical Cognition and Electrophysiology*

- Examined behavioral and posterior P300 responses during mathematical verification in adolescent athletes and controls.
- Conducted EEG/ERP preprocessing and analysis using EEGLAB and ERPLAB.
- Investigated how individual differences in mathematical proficiency account for apparent group differences in neural responses.

Graduate Research Assistant — Child Language Laboratory, NTHU 2023–2025
Advisor: Prof. Hsin-Jen Hsu
Master's Thesis: *Language Lateralization and Reading in ASD*

- Examined developmental differences in language and face lateralization across children with ASD, typically developing children, and typically developing adults.
- Designed experimental paradigms using E-Prime, collected functional transcranial Doppler ultrasound (fTCD) data, and performed cerebral lateralization analyses for 60 participants.
- Revealed distinct patterns of language lateralization between typically developing children and children with ASD, while face processing remained consistently right-lateralized across groups.

Project: *Language Development and Sensorimotor Function in DLD*

- Investigated associations between sensorimotor function and language development in Mandarin-speaking children with developmental language disorder (DLD).
- Administered standardized assessments of intelligence, language, reading, and behavioral functioning and assisted with eye-tracking task implementation and data collection.

Visiting Student Researcher — University of California, Berkeley 2025
Project: *Language Anxiety and Achievement*
Advisor: Prof. Joanna Reed

- Investigated relationships between language anxiety and English language achievement among Mandarin-speaking exchange students.
- Independently designed and completed the project, including research design, survey development, participant recruitment, data collection, statistical analysis, and conference presentation.
- Findings indicated that language proficiency and anxiety did not show a simple inverse relationship, highlighting substantial individual variability in second-language learning outcomes.

Undergraduate Research Assistant — Child Language Laboratory, NTHU 2021-2023
Advisor: Prof. Hsin-Jen Hsu
Project: *Adaptive Digital Learning and Neurophysiological Feedback*

- Administered standardized language and nonverbal cognitive assessments in children.
- Conducted fTCD language tasks involving word generation, listening comprehension, and reading.

Undergraduate Research Assistant — NTHU & NTCU 2020
Advisor: Prof. Kuo-Long Chen & Yuan Yuan
Project: *Numerical Cognition and Arithmetic Processing*

- Investigated how numerical representations and problem-solving strategies influence arithmetic processing.
- Independently designed a PsychoPy-based experimental paradigm and collected behavioral data from over 50 university students.

Honors & Awards

Elected Member, Phi Tau Phi Scholastic Honor Society, Taiwan 2026
Academic Talent Development Fellowship (UC Berkeley), Ministry of Education 2024
International Conference Travel Grant, Ministry of Education 2023
Undergraduate Research Creativity Award, National Science and Technology Council 2022

Teaching Experience

Teaching Assistant — NTHU 2023–2025
Graduate Course

- Research on Theories and Practice of Special Education: supported instruction in research design, experimental methodology, and laboratory orientation in neurophysiological research methods.

Undergraduate Courses

- Teaching Materials and Methods for Children with Special Needs (2 semesters)
- Theory and Practice of Individualized Education Programs (2 semesters)
- Special Education Practicum I & II (3 semesters)

Invited Talks and Workshops

Developing a Research Proposal for Undergraduate Research 2026
College of Education, National Tsing Hua University
Problem Behaviors Across Disability Categories 2024
Teacher Education Center, National Tsing Hua University
Special Education Lesson Planning and Curriculum Design Workshop 2023, 2024
Teacher Education Center, National Tsing Hua University
Introduction and Application of PsychoPy in Experimental Design 2021
Department of Special Education, National Tsing Hua University

Technical Skills

Neuroimaging: fMRI (SPM, FSL, FreeSurfer); EEG/ERP (EEGLAB, ERPLAB); fTCD (DopOSCCI)
Programming and Data Analysis: MATLAB; Python; R; SPSS
Experimental Design: E-Prime; PsychoPy
Educational Measurement and Statistics: Rasch Modeling; Item Response Theory (ConQuest)